

This document details the execution pipeline for the Study Notes Automation suite. Steps 1-3 are linear; step 4 branches per-stage across three parallel implementations, selected via `--stage1-impl/--stage2-impl` (see `readme.md`'s Quick Start).

1 / 3

```

[src/claude_cli_subprocess/stage1_cli.py]
| | |
| | | └─ run_router() →
subprocess.run(["claude", "-p", ...]) ← claude CLI
| | | [Env Stabilization]
| | | └─ Copy files into `AI-Chapter-Notes/<Folder>/` & update `_hold` / SHA-256
state
|
| └─ STAGE 2: Target Selection + Note Generation
[dispatch.generate_notes(impl), src/agents/dispatch.py]
| | [Env Stabilization]
| | └─ Select first folder missing `.docx`, `_notes_done`, `_hold`, or
`_notes_FAILED.txt`
| | [API/CLI Execution] (Mock/no-llm mode: metadata-only check, 0 body
tokens)
| | └─ --stage2-impl legacy (default) → run_generate()
[src/direct_api/stage2_api.py]
| | |
| | | └─ tool-call loop (up to
config.MAX_TURNS) → client.messages.create() ← Anthropic API
| | | └─ --stage2-impl graph ────────────→ run_stage2_chapter()
[src/agents/stage2_graph.py]
| | |
| | | └─ author/figure/compiler nodes
[src/agents/base.py] → client.messages.create() ← Anthropic API
| | | └─ --stage2-impl subprocess ───────────→ run_stage2_chapter()
[src/claude_cli_subprocess/stage2_cli.py]
| | |
| | | └─ run_claude_cli() →
subprocess.run(["claude", "-p", ...]) ← claude CLI
| | | [Env Stabilization]
| | | └─ Confirm `.docx` file written to disk & drop `_notes_done` completion
marker (llm-full mode only)
|
| ▼
5. Return & Exit Handling (System Reset & Retry Handshake)
| | [API Execution / Env Stabilization]
| | └─ On a RETRYABLE API error (rate limit / server overload / network issue --
| | see `classify_api_error()` in func_tools_and_utils.py, which tells these
| | apart from a non-retryable billing/config error):
| | | └─ Pipeline writes the actual retry time (read from the API's response
| | | headers when available) to `<repository-root>\state\retry-epoch.txt`
| | | & exits with code 42
| | | └─ `wsl-study-notes-processor.sh` passes code 42 up to `win-environment-
| | | setup.ps1`
| | | └─ PowerShell registers one-time Windows wake task (`StudyNotesRetry`) for
| | | reset epoch
| | |
| | | └─ On a NON-retryable error (e.g. insufficient API credit): writes
| | | `_notes_FAILED.txt`
| | | with a clear explanation and exits non-zero -- no wake task is scheduled,
| | | since
| | | waiting would not fix it.
| | |
| | | [Env Stabilization]
| | | └─ On Success (Exit Code 0):
| | | | └─ Clean up progress cache file
| | | | └─ If the job FAILED for a non-retry, non-success reason: leave the

```

```
machine
  |   awake (do not sleep) so the failure is visible -- see win-environment-
setup.ps1
  |   ↳ Otherwise: release Win32 keep-awake lock & trigger Modern Standby (turn
off screen)
```